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Top Skills

Python (Programming Language)
Natural Language Processing
Speech Recognition

Languages

Urdu (Native or Bilingual)
English (Full Professional)
Hindi (Native or Bilingual)

Certifications

Miracles of Human Language: An Introduction to Linguistics
Algorithms Specialization
Mathematics for Machine Learning Specialization
Introduction to the Philosophy of Cognitive Sciences
Introduction to Deep Learning (with Honors)

Honors-Awards

Deloitte Innovation Award, Smart India Hackathon

Publications

AutoAnnoMI: A Framework to Automate the Annotations of MI Conversations using LLMs
Beginner's Guide to Audio Data
Training Deep Neural Networks with Batch Normalization
A Fully Generative Motivational Interviewing Counsellor Chatbot for Moving Smokers Towards the Decision to Quit

Zafarullah Mahmood

Research Collaborator @ CAMH | MASc CompEng @ UofT
Toronto, Ontario, Canada

Summary

NLP researcher and engineer with experience delivering real-time NLP applications and scalable ML systems, from model design through to cloud deployment. Recently completed a Master's at the University of Toronto after taking a career break to deepen research expertise. Eligible to work in Canada without sponsorship.

Experience

CAMH

Research Collaborator
September 2023 - August 2025 (2 years)
Toronto, Ontario, Canada

Worked on project MIBot: A Collaboration with researchers, clinicians, humanists and data scientists. MIBot is a fully generative, expert-informed and validated chatbot built on a large language model to deliver Motivational Interviewing therapy. It measurably increases smokers' confidence to quit through text-based conversations.

Dialpad

Natural Language Processing Engineer
April 2019 - August 2023 (4 years 5 months)
Kitchener, Ontario, Canada

: Sentiment Analysis, Natural Language Processing, Text-to-Speech, Inverse Text Normalization, Truecasing, Punctuation Restoration, Language & Acoustic Models, Grapheme-to-Phoneme, BERT, Transformers, N-gram Language Models, Hugging Face transformers, PyTorch, Keras, Kaldi, NeMo, spaCy, NLTK, Kubeflow, Model training, testing and benchmarking

:

- Fine-tuned and deployed a real-time BERT-based English sentiment model for Spanish-English bilingual support while maintaining performance on the original English test set .

- Contributed to the development of an n-gram boosting algorithm that dynamically adjusted decoder probabilities of a speech-to-text model, improving domain-specific keyword recognition by 26%.
- Improved capitalization error by ~40% and production inference time by 4% by replacing a two-stage pipeline for punctuation and casing restoration with a single multi-classification BERT model.
- Trained a transformer-based grapheme-to-phoneme model using Hugging Face Transformers on public and proprietary pronouncing dictionaries and achieved a 70% accuracy improvement over the existing production model.
- Automated data ingestion, retention policy enforcement and de-anonymization workflows and created Kubeflow pipelines for language model training, testing and benchmarking.

Exzeo

Data Scientist

June 2018 - March 2019 (10 months)

New Delhi Area, India

: Speech Diarization, Text and Conversation Analysis, Google DialogFlow, NLP-based Recommendation Systems

:

- Built an AWS-based pipeline to transcribe 5000+ hours of customer service calls and analyzed them using NLTK and spaCy to improve insurance claim assessments.
- Prototyped a chatbot using Google DialogFlow with fallback mechanisms linked to internal databases for better intent matching.
- Deployed an RNN-based text-completion model for mobile devices using TensorFlow Lite to assist insurance adjusters in report writing.

ISRO - Indian Space Research Organization

Software Development Intern

July 2017 - December 2017 (6 months)

Ahmedabad, Gujarat, India

NLP for Indian languages

Education

University of Toronto

Master of Applied Science, Computer Engineering · (2023 - August 2025)

Jamia Millia Islamia

Bachelor of Technology, Computer Engineering · (2014 - 2018)

St. Michael's High School - India

Senior Secondary School Certificate, Natural Sciences + Computer
Science · (2011 - 2014)